60.4

AIME 2024 avg@32

VAPO: Value-Model-Based RL for Advanced Reasoning

Surpassing DAPO by 10+ Points on Long Chain-of-Thought Reasoning

ByteDance Seed | Qwen2.5-32B Base | No SFT Data

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO / DAPO)	Value-Model-Based (VAPO)
• Trajectory-level advantage	• Precise token-level credit assignment
• Uniform assignment to all tokens	• Traces each action's impact on returns
• High variance Monte Carlo estimates	• Lower-variance per-token estimates
• Subtle step errors cause catastrophic failures undetected	• Detects subtle step-level errors early
• Poor generalization across samples	• Better generalization and sample efficiency

Insight: Value models are critical when subtle step-level errors cause catastrophic failures in multi-step reasoning.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

01

Value Model Bias

Long trajectories and bootstrapped learning introduce severe bias in value estimates. The value model struggles to propagate sparse rewards back through thousands of reasoning tokens.

02

Heterogeneous Sequence Lengths

Different response lengths require different bias-variance tradeoffs. A single fixed GAE lambda cannot optimally serve both short and long reasoning chains.

03

Reward Signal Sparsity

Only the final <eos> token receives reward. With long CoT, the signal is extremely sparse, making credit assignment and value learning inherently difficult.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Challenge	Technique	Description
Value Model Bias	Value-Pretraining	Pre-trains value model before RL to reduce initial bias
Heterogeneous Lengths	Decoupled-GAE	Separates GAE computation for more robust value estimation
Reward Sparsity	Length-Adaptive GAE (novel)	Dynamically adjusts λ based on response length
Exploration	Clip-Higher	Asymmetric clipping to encourage exploration
Variance Reduction	Token-level Loss	Normalizes loss at token level vs sequence level
	Positive Example LM Loss	Self-imitation learning on correct trajectories

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Short Response

Lower λ

→ Lower variance, higher bias acceptable
→ Rely more on value model bootstrap

Long Response

Higher λ

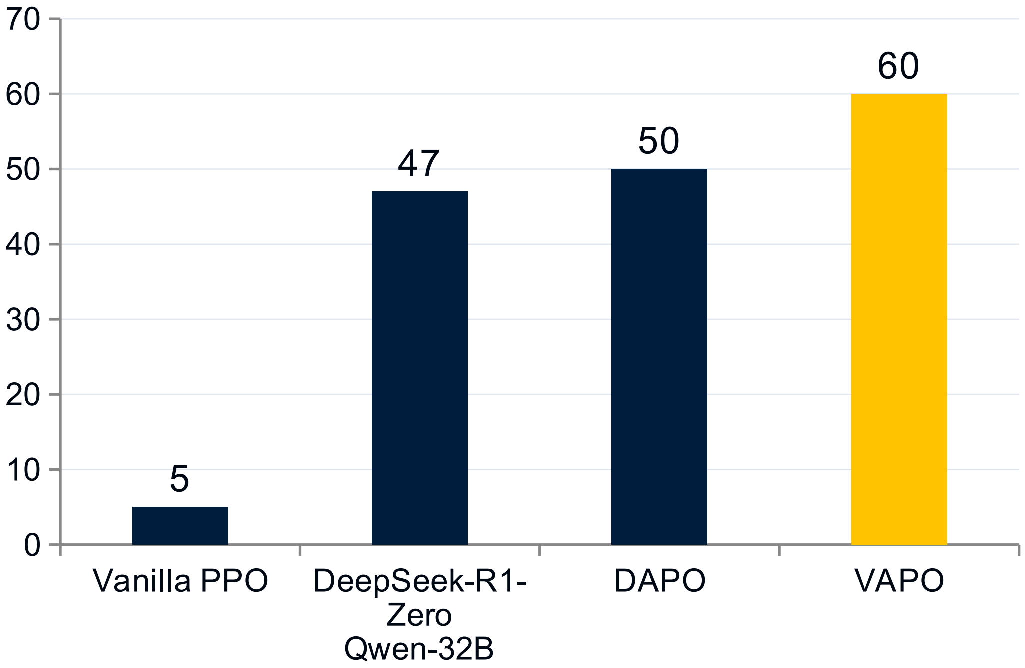
→ Lower bias, higher variance tolerated
→ Rely more on actual returns (Monte Carlo)

The Bias-Variance Tradeoff

- GAE(λ) interpolates between biased value estimates (λ = 0) and high-variance Monte Carlo returns (λ = 1).
- A single fixed λ cannot optimally serve both short and long reasoning chains.
- Length-Adaptive GAE dynamically selects λ based on response length.
- Removing it drops AIME score from 60 to 45 … contribution.

Novel Contribution

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



+10.4
points over DAPO

+13
points over DeepSeek-R1-Zero

12x
improvement over Vanilla PPO

Training curve (figure_2.jpg): VAPO reaches ~60% at step 4,500 vs DAPO ~50% at step 5,500

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

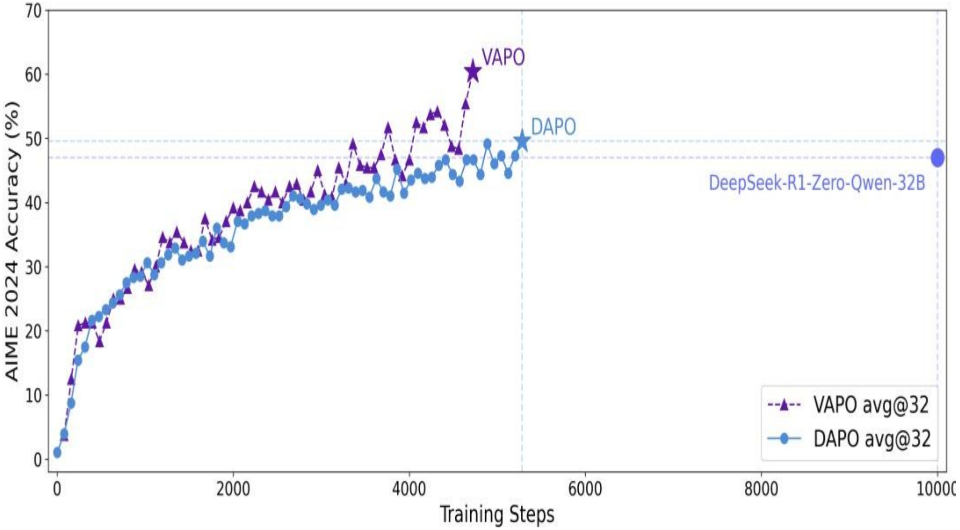


Figure 2: VAPO vs DAPO training curves on AIME 2024

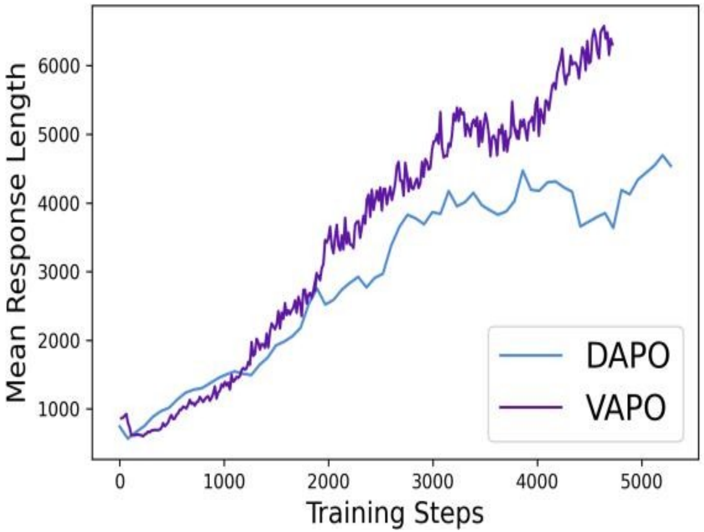


Figure 3: Mean response length over training

~5,000

Steps to SOTA

0

Training crashes

~6,000

Tokens per response

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME 2024 Score	Δ from VAPO
VAPO (full)	**60**	**—**
w/o Group-Sampling	55	-5
w/o Positive Example LM Loss	54	-6
w/o Token-level Loss	53	-7
w/o Clip-Higher	46	-14
w/o Length-adaptive GAE	45	-15
w/o Decoupled-GAE	33	-27
w/o Value-Pretraining	**11**	**-49**

Key Insights

- Value-Pretraining is the single most critical component — removing it drops score from 60 to 11.
- Without it, value-model-based RL barely works.
- Decoupled-GAE is the second most important (drop to 33).
- Length-adaptive GAE contributes 15 points, validating the hypothesis.

Value-Pretraining: +49 pts

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1. Value-model-based methods can surpass value-model-free when core challenges are systematically addressed.
2. Length-Adaptive GAE is critical for heterogeneous sequence lengths — contributes 15 points.
3. Value-Pretraining is the single most important component — without it, score drops from 60 to 11.
4. VAPO is highly stable: zero training crashes across multiple independent runs.
5. Reaches SOTA in ~5,000 steps with longer, more elaborate reasoning chains (~6,000 tokens).

Implication: Value-model-based RL is a viable and superior path for complex reasoning — if bias, length heterogeneity, and sparsity are handled with care.